

SOFTWARE ENGINEERING LAB – LAB 3

Component Modelling & Architectural Pattern Selection

Patient Health Record Consent Management System

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1. Architecture Selection

We chose Layered Architecture for the Patient Health Record Consent Management System. It separates user interaction, authentication/authorization, consent management, health-record access, notifications, auditing, and data storage into clearly defined responsibilities.

2. Architectural Style Analysis

Architecture	Advantages	Disadvantages	Suitability
Layered	Clear separation; easier maintenance; structured design.	Some layer overhead; dependencies between layers.	Best suited
Microservices	Independent deployment/scaling; fault isolation.	Operational complexity; network latency; consistency challenges.	More complex
Client-Server	Centralized control; simple deployment; consistent data.	Single point of failure; scalability bottleneck.	Possible, less suitable

3. Component Identification

- 1. Patient / Clinic Portal:** Patient/clinic interaction, consent actions and access requests.
- 2. Authentication & Authorization Service:** Authentication, role verification, access control and authorization.
- 3. Consent Management Service:** Time-bound consent, revocation, expiry and consent validation
- 4. Health Record Service:** Retrieves and provides medical/diagnostic records to authorized users.
- 5. Patient Health Records Database:** Stores patient medical history and diagnostic records.
- 6. Audit Trail Service:** Records consent grants, revocations and record access in an append-only trail.
- 7. Notification Service:** Sends consent-expiry and access-related notifications.

4. Interfaces and Component Interactions

Interface	Components	Technology
Authentication API	Portal ↔ Authentication & Authorization	HTTPS/REST
Consent Management API	Portal ↔ Consent Management	HTTPS/REST
Authorization Check	Authentication & Authorization ↔ Consent Management	HTTPS/REST
Notification API	Consent Management ↔ Notification	HTTPS/REST
Record Access API	Consent Management ↔ Health Record	HTTPS/REST
Audit Logging API	Consent Management ↔ Audit Trail	HTTPS/REST
Database Query	Health Record ↔ Patient Records DB	SQL/JDBC
Audit Log Write	Audit Trail ↔ Audit Log DB	SQL/JDBC

5. Security Advantage

Authentication & Authorization is separated from the Health Record Service, with Consent Management between users and protected records. Identity, role and valid consent can be checked before access. Consent grants, revocations and access events are recorded through the Audit Trail Service.

6. Performance Benefit

Consent Management can validate consent and authorization before requesting medical records, avoiding unnecessary database access and keeping the main consent/access flow efficient.

7. Conclusion

Layered Architecture provides a clear and maintainable structure. It separates authentication, consent management, record access, notification, auditing and data storage while supporting controlled and auditable access to patient records.